Genomic Clustering and Survival Analysis in Cancer: A Data-Driven Approach to Identifying Metastatic Risk Subtypes

[Website link - Click here]

[Github Repo link - Click here]

Note: Full model training and bootstrapping (100 iterations) requires ~50 minutes. The bootstrapping block has been disabled for convenience; the current script takes roughly **25 minutes** to render. Please refer to the GitHub repository for the pre-calculated `bootstrapping_results.csv` and the final `Rendered_Report`

Reproducibility Instructions

To render this document locally:

1. Install Quarto and R
2. Install required packages:

```
# Installing the required packages

packages <- c(
  "dplyr", "tidyr", "broom", "knitr", "data.table",
  "survival", "randomForestSRC", "cluster", "corrplot", "ggplot2",
  "pec", "survminer", "fpc","tibble","pheatmap"
)

# only install the missing packages
install.packages(setdiff(packages, rownames(installed.packages())))
```

3. Run locally:

   - Open a terminal (e.g., RStudio Terminal, Git Bash, or Command Prompt).

   - Navigate to the folder containing `Source_code.qmd`

- Then run the following command

```
quarto render Source_code.qmd
```

The rendered output (e.g., HTML) will be created in the same directory.

Note : Since this code file trains 1000 trees RSF model and again 100 bootstrapping clustering it will take approximately around 45 minutes to generate the Output HTML file

```
getwd()
```

```
[1] "D:/Myproject"
```

```
# Load the required libraries

suppressWarnings({
  library(dplyr)
  library(tidyr)
  library(broom)
  library(knitr)
  library(data.table)
  library(survival)
  library(randomForestSRC)
  library(cluster)
  library(corrplot)
  library(pec)
  library(survminer)
  library(fpc)
  library(ggplot2)
  library(tibble)
  library(pheatmap)
})

# R version
R.version.string
```

```
[1] "R version 4.4.2 (2024-10-31 ucrt)"
```

```r
# Package Information
pkgs <- c(
  "fpc", "survminer", "ggplot2", "pec",
  "corrplot", "cluster", "randomForestSRC",
  "survival", "data.table", "knitr", "broom",
  "tidyr", "dplyr"
)

df_p <- data.frame(
  #Package = pkgs,
  Version = sapply(pkgs, function(x) as.character(packageVersion(x)))
)
kable(df_p, caption = "R Package Versions Used in Analysis")
```

R Package Versions Used in Analysis

	Version
fpc	2.2.14
survminer	0.5.2
ggplot2	4.0.1
pec	2025.6.24
corrplot	0.95
cluster	2.1.8.2
randomForestSRC	3.5.1
survival	3.8.6
data.table	1.17.8
knitr	1.50
broom	1.0.10
tidyr	1.3.1

	Version
dplyr	1.1.4

```
# Loading the data directly from Github

url_clipat <- paste("https://media.githubusercontent.com/media/cBioPortal/datahub",
                    "/refs/heads/master/public/msk_chord_2024/data_clinical_patient.txt",
                    sep="")
url_clisamp <- paste("https://media.githubusercontent.com/media/cBioPortal/datahub",
                     "/refs/heads/master/public/msk_chord_2024/data_clinical_sample.txt",
                     sep="")

url_timediag <- paste("https://media.githubusercontent.com/media/cBioPortal/datahub/refs/heads/master/public/msk_


url_time_progress<- paste("https://media.githubusercontent.com/media/cBioPortal/datahub/refs/heads/master/public/

url_mutations <- paste("https://media.githubusercontent.com/media/cBioPortal/datahub/refs/heads/master/public/msk

url_cna <- "https://github.com/cBioPortal/datahub/raw/master/public/msk_chord_2024/data_cna_hg19.seg"

download.file(url_cna, "data_cna.seg", mode = "wb", quiet=TRUE)

url_prior_meds <- paste("https://media.githubusercontent.com/media/cBioPortal/datahub/refs/heads/master/public/ms
```

Background

Metastasis remains the leading cause of cancer mortality, yet outcomes vary widely even within cancer type. Traditional subtyping often fails to capture non-linear interactions between multi-omic features and longitudinal survival. Here, we present a framework that uses a supervised survival model to learn a patient distance metric, then clusters on that representation to identify metastatic-risk subtypes.

Dataset and Data Cleaning

This study utilized the MSK-CHORD clinicogenomic dataset, which contains genomic and clinical information from over 25,000 tumor samples. To define a cohort suitable for metastasis risk analysis, patients were excluded if they (1) had documented metastatic progression prior to or at sequencing, (2) had Stage IV or distant metastatic disease at baseline, (3) had progression recorded at or before sequencing, (4) had no post-sequencing follow-up time, or (5) had incomplete genomic or outcome data. Eligible patients were identified by integrating multiple clinical datasets, including data_clinical_patient.txt, data_timeline_diagnosis.txt, and data_timeline_progression.txt. Genomic features are retained only from primary tumor samples and eligible patients and aggregated at the patient level. Gene-level mutation indicators were derived from data_mutations.txt, tumor mutational burden (TMB) and cancer type from data_clinical_sample.txt, and global copy number alteration (CNA) burden was calculated from the segmentation file data_cnahg19.seg. Clinical timeline data from data_timeline_progression.txt were used to construct a time-to-metastasis (TTM) outcome measured from the date of genomic sequencing (time 0). Eligible patients were followed from sequencing until first documented metastasis or last recorded follow-up.

```
# Reading the clinical patient data

clinical_patient <- read.delim(url_clipat, comment.char="#",
                               stringsAsFactors = TRUE)




#### clinical_patient subset
clinical_features <- clinical_patient %>%
  dplyr::select(PATIENT_ID,
         GENDER,
         RACE,
         ETHNICITY,
         CURRENT_AGE_DEID,
         SMOKING_PREDICTIONS_3_CLASSES,
         OS_MONTHS,
         OS_STATUS)
#clinical_features
```

```r
# reading Clinical_sample

clinical_sample <- read.delim(url_clisamp, comment.char="#",stringsAsFactors = TRUE)

#View(clinical_sample)
#kable(table(clinical_sample$CANCER_TYPE))
#names (clinical_sample)
#length(unique(clinical_sample$PATIENT_ID))
#nrow(clinical_sample)
#typeof(clinical_sample)
#table(clinical_sample$SAMPLE_TYPE)
#str(clinical_sample)
#clinical_sample[clinical_sample$PATIENT_ID == "P-0000012",]
#clinical_sample[clinical_sample$PATIENT_ID == "P-0000825",]
#clinical_sample[clinical_sample$PATIENT_ID == "P-0002880",]
#clinical_sample[clinical_sample$SAMPLE_TYPE == "Primary", ]
### Viewing multiple samples per patient
#sample_counts <- clinical_sample %>% count(PATIENT_ID, name = "n_samples")
#sample_counts[sample_counts$n_samples == 2,]
#View(sample_counts)

kable(
  as.data.frame(table(clinical_sample$CANCER_TYPE)) |>
    setNames(c("Cancer type", "Freq"))
)
```

Cancer type	Freq
Breast Cancer	5368
Colorectal Cancer	5543
Non-Small Cell Lung Cancer	7809
Pancreatic Cancer	3109
Prostate Cancer	3211

```r
# Reading Timeline Diagnosis

timeline_diag <- read.delim(url_timediag, comment.char="#",stringsAsFactors = TRUE)


#If timeline_diag$START_DATE > 0 represents new primary diagnosis after sequencing, then:
#That is clinical information evolving over time —but it does NOT affect survival baseline definition.

#timeline_diag %>%
  #filter(START_DATE > 0)

#checking
timeline_diag$SUMMARY <- trimws(timeline_diag$SUMMARY)
#timeline_diag[timeline_diag$PATIENT_ID == "P-0000612", ]

#Determine Baseline Stage at Sequencing & Remove Patients Metastatic at Baseline
non_met_baseline <- timeline_diag %>%
  filter(START_DATE <= 0) %>% # all diagnosis records happened before or at genomic
  group_by(PATIENT_ID) %>%
  slice_max(START_DATE) %>% #What was the Patient's stage at the time of seqencing?
  filter(STAGE_CDM_DERIVED != "Stage 4" ) %>%
  filter(!SUMMARY %in% c("Distant", "Distant metastases/systemic disease"))



#at-risk cohort

#n_distinct(non_met_baseline$PATIENT_ID)

#checkpoint = 14080
```

```r
#timeline_progress <-  read.delim("msk_chord_2024_data/data_timeline_progression.txt", header = TRUE, sep = "\t",

timeline_progress <-  read.delim(url_time_progress, comment.char="#",stringsAsFactors = TRUE)
```

```r
# patients who cannot be included in a time-to-metastasis analysis because they
## already had metastasis at the time of sequencing.
pre_baseline_met <- timeline_progress %>%
  filter(
    START_DATE <= 0,
    PROGRESSION == "Y"
  ) %>%
  distinct(PATIENT_ID)

## eligible patients from clinical_patient_data

eligible_patients <- clinical_features %>%
  filter(PATIENT_ID %in% non_met_baseline$PATIENT_ID) %>%
  # keep only non-metastatic at sequencing
  filter(!PATIENT_ID %in% pre_baseline_met$PATIENT_ID)
  # remove anyone who had metastasis recorded pre-baseline

#nrow(eligible_patients) #9370

# Extracting post-baseline metastasis events

met_events <- timeline_progress %>%
  filter(START_DATE > 0,
         PROGRESSION == "Y") %>%
  group_by(PATIENT_ID) %>%
  summarise(first_met_day = min(START_DATE),
            .groups = "drop")
#View(met_events)
#nrow(met_events)

#===================================================================
# This block is not needed as Start time will not be included in the RSF model
#Extracting Baseline(sequencing time)
start_time <- timeline_diag %>%
  #filter(PATIENT_ID %in% met_events$PATIENT_ID)%>%
  dplyr::select(PATIENT_ID, START_DATE) %>%
```

```r
  group_by(PATIENT_ID) %>%
  slice_max(START_DATE, with_ties = FALSE)
#start_time[start_time$PATIENT_ID == "P-0000612", ]

#View(start_time)
#nrow(start_time)
#n_distinct(start_time$PATIENT_ID)
#start_time[start_time$START_DATE == 0,]
#start_time[start_time$PATIENT_ID ==  "P-0003523" , ]
#======================================================================



# Extarcting  last follow-up time
last_followup <- timeline_progress %>%
  group_by(PATIENT_ID) %>%
  summarise(last_day = max(START_DATE, na.rm = TRUE),
            .groups = "drop")

#View(last_followup)



#==========================================================================
## Time to metastasis event

ttm_data <- eligible_patients %>%
  #left_join(start_time, by ="PATIENT_ID" )%>%
  left_join(met_events, by = "PATIENT_ID") %>%
  left_join(last_followup, by = "PATIENT_ID") %>%
  mutate(
    event = ifelse(!is.na(first_met_day), 1, 0),
    time  = ifelse(event == 1,
                 first_met_day,
                 last_day)
  ) %>%
  filter(!is.na(time),
       time > 0)   #The latest recorded event for this patient happened before                        #sequencin
                     #This patient cannot contribute meaningful survival info
```

```r
#View(ttm_data)
#summary(ttm_data$time)
#kable(table(ttm_data$event))
#names(ttm_data)

#sum(ttm_data$time ==  0)
#nrow(ttm_data)
#n_distinct(ttm_data$PATIENT_ID)
#ttm_data[ttm_data$PATIENT_ID == "P-0000612", ]
#================================================================================

analysis_df <- ttm_data %>%
dplyr::select(PATIENT_ID,first_met_day,last_day, time, event)

## START_DATE = Sequencing day
## first_met_day == first metastasis event
## last_day = last_day_followup
## event = metastasis 1/0
## time = if metastasis =1 then time = first_met_day; metastasis =0 then time = last_day_followup
#head(analysis_df)
#names(analysis_df)
nrow(analysis_df) # 7930
```

```
[1] 7930
```

```r
n_distinct(analysis_df$PATIENT_ID) # 7930
```

```
[1] 7930
```

```r
kable(
  as.data.frame(table(analysis_df$event)) |>
    setNames(c("Metastasis", "Freq"))
)
```

Metastasis	Freq
0	4401
1	3529

```
### TMB Block



### Primary sample = baseline sequencing
## Filter only primary sample type in clinical_sample

baseline_tmb_check <- clinical_sample %>%
  filter(SAMPLE_TYPE == "Primary")%>%
  dplyr::select(SAMPLE_ID,PATIENT_ID, SAMPLE_TYPE,  PRIMARY_SITE, TMB_NONSYNONYMOUS)
## Patients with 2 primary tumors
#baseline_tmb_check[baseline_tmb_check$PATIENT_ID == "P-0002880",]
#baseline_tmb_check[baseline_tmb_check$PATIENT_ID == "P-0004121",]
#baseline_tmb_check[baseline_tmb_check$PATIENT_ID == "P-0005813",]

## patient with 1 primary and 1 metastaic tumor sampled

#clinical_sample[clinical_sample$PATIENT_ID== "P-0000012",]

### tmb df = ## Filter only primary sample type in clinical_sample
## if there are 2 primary tumor samples pull the first sampled tumor's tmb

baseline_tmb <- clinical_sample %>%
  filter(SAMPLE_TYPE == "Primary") %>%
  group_by(PATIENT_ID)%>%
  slice_head(n = 1) %>% #pull the first sampled tumor's tmb
  ungroup() %>%
  dplyr::select(SAMPLE_ID, PATIENT_ID, TMB_NONSYNONYMOUS)

tmb_df <- baseline_tmb  %>%
  filter(PATIENT_ID %in% ttm_data$PATIENT_ID)
```

```r
nrow(tmb_df) #6655
```

```
[1] 6655
```

```r
#length(tmb_df$PATIENT_ID)
#length(unique(tmb_df$PATIENT_ID))


#baseline_tmb[baseline_tmb$PATIENT_ID == "P-0000066", ]
```

```r
### Mutations Block

#mutations <- read.delim("msk_chord_2024_data/data_mutations.txt", header = TRUE, sep = "\t", stringsAsFactors =

mutations <- read.delim(url_mutations, comment.char="#",
                               stringsAsFactors = TRUE)
#View(mutations)
#names(mutations)
#unique(mutations$Variant_Classification)
#mutations[mutations$Tumor_Sample_Barcode == "P-0000012-T03-IM3",]
#mutations[mutations$Tumor_Sample_Barcode == "P-0000012-T02-IM3",]

## Mapping tumor barcode sample ID to patient ID byTumor_Sample_Barcode

mutations_clini_merged <- mutations %>%
  left_join(clinical_sample %>% dplyr:: select(Tumor_Sample_Barcode = SAMPLE_ID, PATIENT_ID, SAMPLE_TYPE ),
            by = "Tumor_Sample_Barcode")%>%
  filter(Tumor_Sample_Barcode %in% baseline_tmb$SAMPLE_ID) # only keep tumor sample sampled from primary tumor
#unique(mutations_clini_merged$SAMPLE_TYPE) ## check haiving only primary tumor type
#length(unique(mutations_clini_merged$Tumor_Sample_Barcode))

#length(unique(mutations_clini_merged$Hugo_Symbol))
#names(mutations_clini_merged)
#nrow ((mutations_clini_merged))
```

```r
# Filter mutations for eligible patients
mutations_eligible <- mutations_clini_merged %>%
  filter(PATIENT_ID %in% ttm_data$PATIENT_ID)
#nrow(mutations_eligible)


## keep mutations that change the protein sequence - nonsynonymous mutations
nonsynonymous_classes <- c(
  "Missense_Mutation",
  "Nonsense_Mutation",
  "Frame_Shift_Del",
  "Frame_Shift_Ins",
  "frameshift_insertion",
  "In_Frame_Del",
  "In_Frame_Ins",
  "Splice_Site",
  "Translation_Start_Site",
  "Nonstop_Mutation",
  "nonsynonymous_SNV"
)


mutations_eligible_ns <- mutations_eligible %>%
  filter(Variant_Classification %in% nonsynonymous_classes)

#mutations_eligible_ns


# selecting top 50 mutated genes
top_genes <- mutations_eligible_ns %>%
  count(Hugo_Symbol, sort = TRUE) %>%
  slice_head(n = 50) %>%
  pull(Hugo_Symbol)
#top_genes

### Created a Binary Matrix
```

```
mutation_matrix <- mutations_eligible_ns %>%
  filter(Hugo_Symbol %in% top_genes) %>%
  distinct(PATIENT_ID, Hugo_Symbol ) %>% #one row per PATIENT_ID.
  mutate(value = 1) %>%
  pivot_wider(
    names_from = Hugo_Symbol,
    values_from = value,
    values_fill = 0
  )
#View(mutation_matrix)
nrow(mutation_matrix) #5878
```

```
[1] 5878
```

```
#names(mutation_matrix)
```

```
## Copy number alterations


#cna_seg <- fread("msk_chord_2024_data/data_cna_hg19.seg")


cna_seg <- fread("data_cna.seg")
#head(cna_seg)
#View(cna_seg)
#names(cna_seg)


#Mapping ID to Patient ID
cna_seg_mapped <- cna_seg %>%
  left_join(clinical_sample %>%
              dplyr::select(SAMPLE_ID, PATIENT_ID),
            by = c("ID" = "SAMPLE_ID"))

# Filter mutations for eligible patients
```

```r
cna_eligible <- cna_seg_mapped %>%
  filter(ID %in% baseline_tmb$SAMPLE_ID)%>% # pimary cna load
  filter(PATIENT_ID %in% ttm_data$PATIENT_ID) # filtering eligible patients


## Global cna burden

cna_burden <- cna_eligible %>%
  mutate(
    segment_length = loc.end - loc.start,
    altered = abs(seg.mean) > 0.3
  ) %>%
  group_by(ID, PATIENT_ID) %>%
  summarise(
    total_length = sum(segment_length),
    altered_length = sum(segment_length[altered]),
    global_cna_burden = altered_length / total_length,
    .groups = "drop")

cna_patients <- cna_burden %>%
  dplyr::select (PATIENT_ID, global_cna_burden)

#nrow(cna_burden)
#length(unique(cna_burden$PATIENT_ID))
#length(unique(cna_burden$ID))
```

```r
# Extracting Cancer type info from Clinical_sample

patient_cancer_type <- clinical_sample %>%
  group_by(PATIENT_ID) %>%
  arrange(SAMPLE_ID) %>%
  slice(1) %>%   # keeps the first row per patient
  ungroup() %>%
  select(PATIENT_ID, CANCER_TYPE)
```

```r
###  genomic features by patient ID

genomic_features <- analysis_df %>%
  dplyr::select(PATIENT_ID,time, event )%>%
  left_join(patient_cancer_type, by = "PATIENT_ID")%>%
  left_join(mutation_matrix, by = "PATIENT_ID") %>%
  left_join(tmb_df, by = "PATIENT_ID") %>%
  left_join(cna_patients, by = "PATIENT_ID")
genomic_features <- genomic_features %>%
  mutate(across(where(is.numeric), ~replace_na(.x, 0)))
#View(analysis_df)
table(genomic_features$CANCER_TYPE)
```

```
         Breast Cancer       Colorectal Cancer
                  2106                    1912
Non-Small Cell Lung Cancer   Pancreatic Cancer
                  2257                     999
       Prostate Cancer
                   656
```

```r
genomic_features$CANCER_TYPE <- as.factor(genomic_features$CANCER_TYPE)
kable(table(genomic_features$event)) # 0: 4401; 1:3529
```

Var1	Freq
0	4401
1	3529

```r
###=================Final table for RFS analysis====================
#View(genomic_features)
#names(genomic_features)

#nrow(genomic_features)
```

```
#n_distinct(analysis_df$PATIENT_ID)
#genomic_features[genomic_features$time == 0,]

####=====================================================================
```

Methods

In this study, RSF was applied to integrate cancer type and genomic features for unsupervised patient stratification. Predictors included gene-level mutations, tumor mutational burden, copy number alteration burden, and cancer type. The RSF model was implemented using the randomForestSRC package in R, allowing stable estimation of survival relationships within the data. The RSF model with 1,000 trees was trained, where each tree was built using a bootstrap sample of patients to improve model stability and robustness.

For every tree, approximately one-third of patients were excluded from the bootstrap sample and served as out-of-bag (OOB) observations, which were used to obtain unbiased error estimates and to compute patient similarity. A patient-level proximity matrix was then constructed by measuring how frequently pairs of patients appeared in the same terminal nodes across trees, with higher proximity indicating more similar survival-associated genomic patterns. This proximity matrix was converted to a distance matrix (1 – proximity) and used for clustering by partitioning around medoids (PAM).The optimal cluster number (K=5) was selected via silhouette analysis and validated by bootstrapping (mean Jaccard index = 0.918). Cluster-level survival differences were assessed with Kaplan-Meier curves and quantified with Cox regression, and variable importance (VIMP) was used to identify predictive genomic features

Results

```
#RSF Clustering with Cancertype and genomic features


# Create survival object
surv_obj <- Surv(time = genomic_features$time, event = genomic_features$event)

# Prepare data for RSF
rsf_data_type <- genomic_features %>%
```

```r
  dplyr::select(-PATIENT_ID, -SAMPLE_ID)

set.seed(123)

# Fit RSF model
rsf_model_type <- rfsrc(
  Surv(time, event) ~ .,
  data = rsf_data_type,
  ntree = 1000,
  proximity = "oob",
  importance = "permute"
)

# Proximity matrix
prox_matrix1 <- rsf_model_type$proximity

# Create distance matrix
dist_matrix1 <- 1 - prox_matrix1
dist_obj1 <- as.dist(dist_matrix1)
```

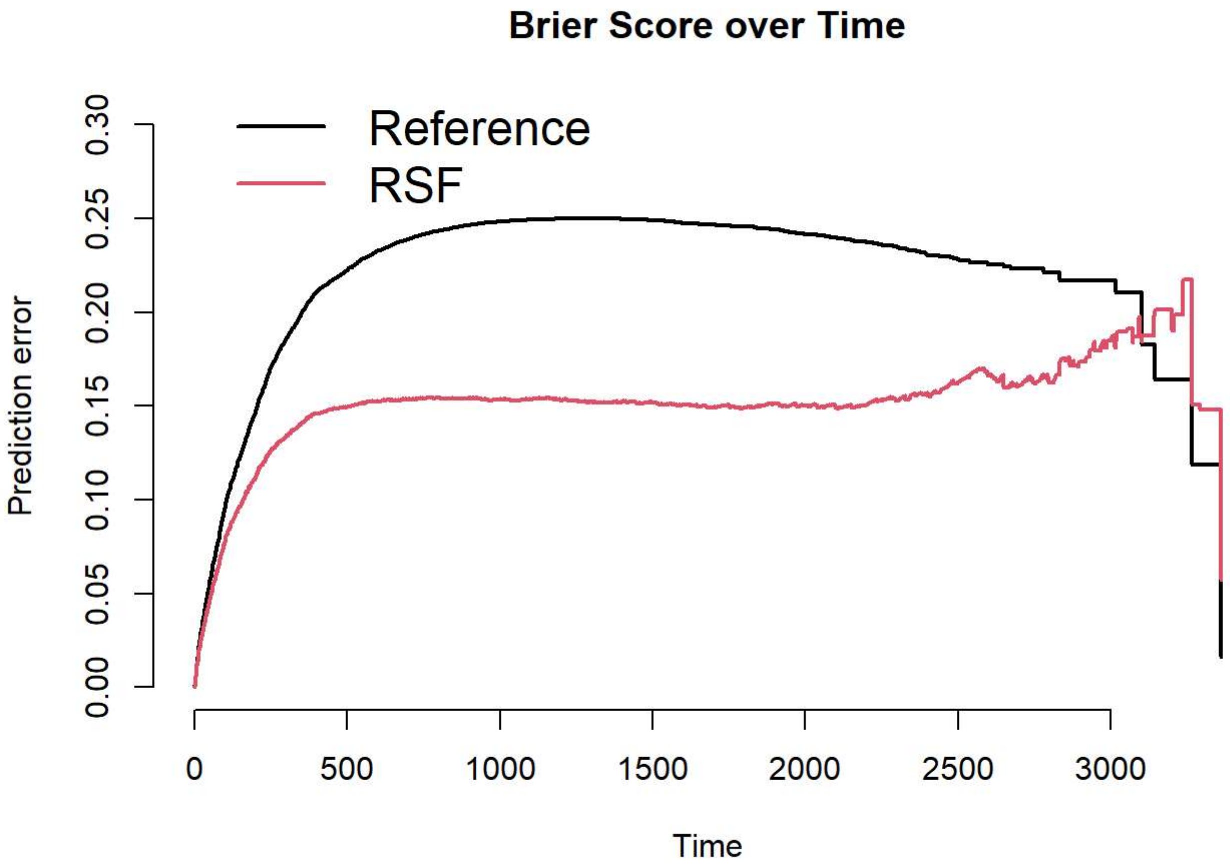
```r
##RSF Model Evaluation


#library(timeROC)


# --- 1. C-index ---
c_index <- concordance(
  Surv(time, event) ~ I(-rsf_model_type$predicted),#Higher predicted values #correspond to higher risk (or were i
  data = rsf_data_type
)$concordance
cat("C-index:", round(c_index, 3), "\n")
```

```
C-index: 0.769
```

```r
# --- 2. Brier Score ---
brier <- pec(
  object = list("RSF" = rsf_model_type),
  formula = Surv(time, event) ~ 1,
  data = rsf_data_type,
  cens.model = "marginal"
)
print(crps(brier))  # Integrated Brier Score
```

```
Integrated Brier score (crps):

          IBS[0;time=3361)
Reference            0.218
RSF                  0.152
```

```r
plot(brier, main = "Brier Score over Time")
title("Brier Score over Time")
```




```
# --- 3. Time-dependent AUC ---=====================================================
#rsf_pred <- rsf_model_type$predicted  # verify interpretation!

#time_points <- c(12, 24, 36)

#roc <- timeROC(
  #T = rsf_data_type$time,
  #delta = rsf_data_type$event,
  #marker = rsf_pred,
  #cause = 1,
```

```
  #times = time_points,
  #iid = TRUE
#)

#for (i in 1:length(time_points)) {
#  cat("AUC at", time_points[i], ":", round(roc$AUC[i], 3), "\n")
#}
```

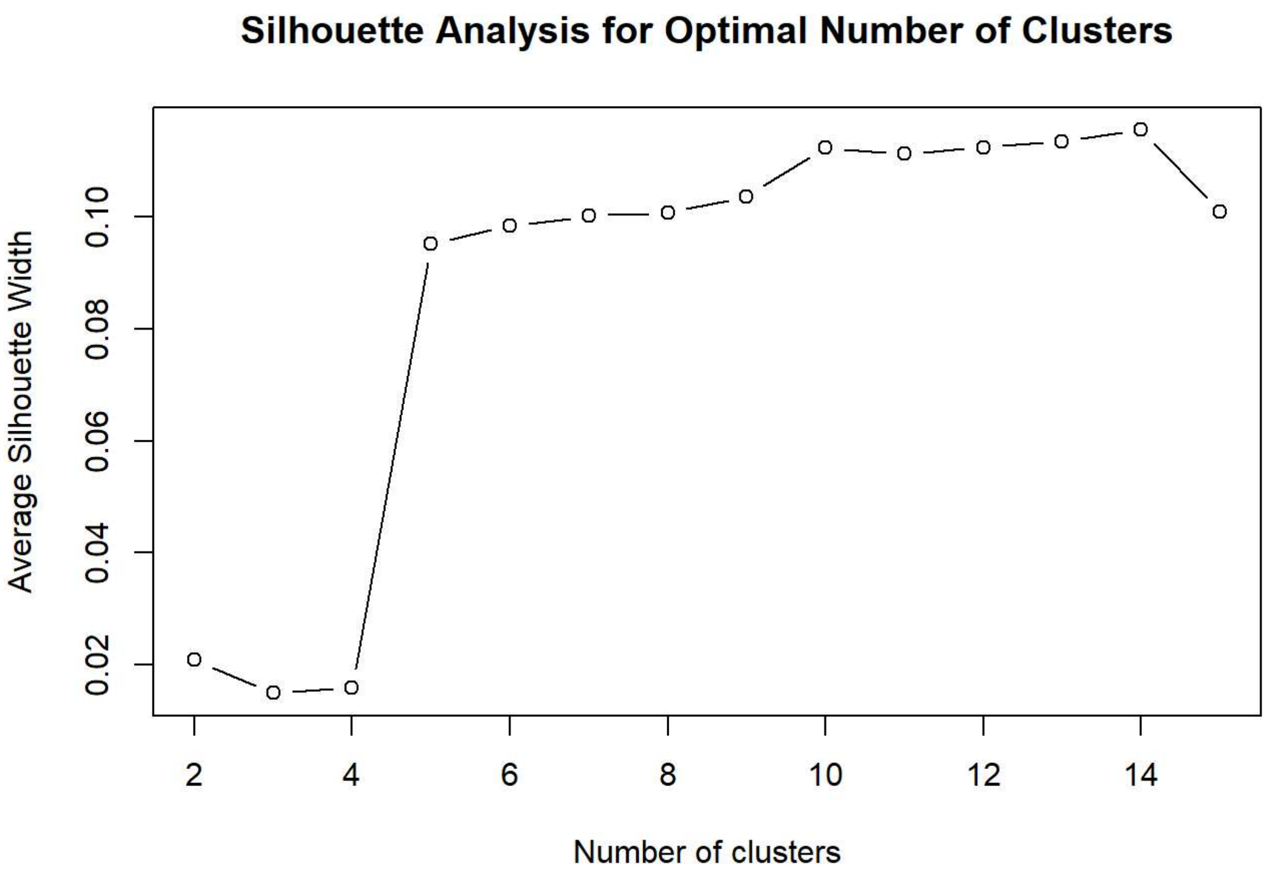
RSF Model Performance and Cluster Stability

The Random Survival Forest (RSF) model demonstrated strong predictive performance for survival outcomes. The model achieved a concordance index (C-index) of **0.769**, indicating good discrimination in ranking patients by risk.

Model performance was further evaluated using the Integrated Brier Score (IBS) over a follow-up time horizon of 3361 days. The RSF model showed improved predictive accuracy compared to the reference model, with an IBS of **0.152**, compared to **0.218** for the reference approach. This reduction in prediction error indicates better overall calibration and accuracy of survival probability estimates.

Overall, these results suggest that the RSF model provides robust predictive performance for time-to-event outcomes, with improved discrimination and calibration relative to the baseline model.

```
# Determine optimal number of clusters (Silhouette method)
sil_width1 <- c()
for (k in 2:15) {
  pam_fit1 <- pam(dist_obj1, k = k, diss = TRUE)
  sil_width1[k] <- pam_fit1$silinfo$avg.width
}

plot(2:15, sil_width1[2:15], type = "b",
     xlab = "Number of clusters",
     ylab = "Average Silhouette Width")

title("Silhouette Analysis for Optimal Number of Clusters")
```




```
###Cluster Stabilty Analysis


# RSF distance matrix
dist_matrix2 <- as.dist(1 - prox_matrix1)

stability_results <- list()

for (k in 2:10) {
  capture.output({
```

```r
    boot <- clusterboot(
      dist_matrix2,
      B = 100,
      bootmethod = "boot",
      clustermethod = disthclustCBI,
      method = "ward.D2",
      k = k,
      seed = 42
    )
  }, file = nullfile())

  stability_results[[k]] <- data.frame(
    k = k,
    mean_jaccard = mean(boot$bootmean),
    min_jaccard  = min(boot$bootmean),
    n_stable     = sum(boot$bootmean > 0.75)
  )
}

stability_df <- do.call(rbind, stability_results)
rownames(stability_df) <- NULL  # clean up row names from rbind
print(stability_df)
```

```
   k mean_jaccard min_jaccard n_stable
1  2    0.9943129   0.9905026        2
2  3    0.9924233   0.9791466        3
3  4    0.8985529   0.6541313        3
4  5    0.9181776   0.6944601        4
5  6    0.7737021   0.3186899        3
6  7    0.8452972   0.6847714        4
7  8    0.8370233   0.5272080        5
8  9    0.7801358   0.3555775        5
9 10    0.7466024   0.4747571        5
```

Cluster stability analysis using 100 bootstrap iterations confirmed that K=5 provided a highly stable solution (Mean Jaccard Index = **0.918**), whereas higher cluster counts (e.g., K=6 or 10) resulted in a significant reduction in stability and the emergence of poorly defined subgroups. Thus the optimal cluster number (K=5) was selected.

```
#Final Clustering (PAM)

# selected k=5 based on stablity analysis
pam_final <- pam(dist_obj1,
                 k = 5,# Based on stability analysis
                 diss = TRUE)

genomic_features$cluster <- factor(pam_final$clustering)
```

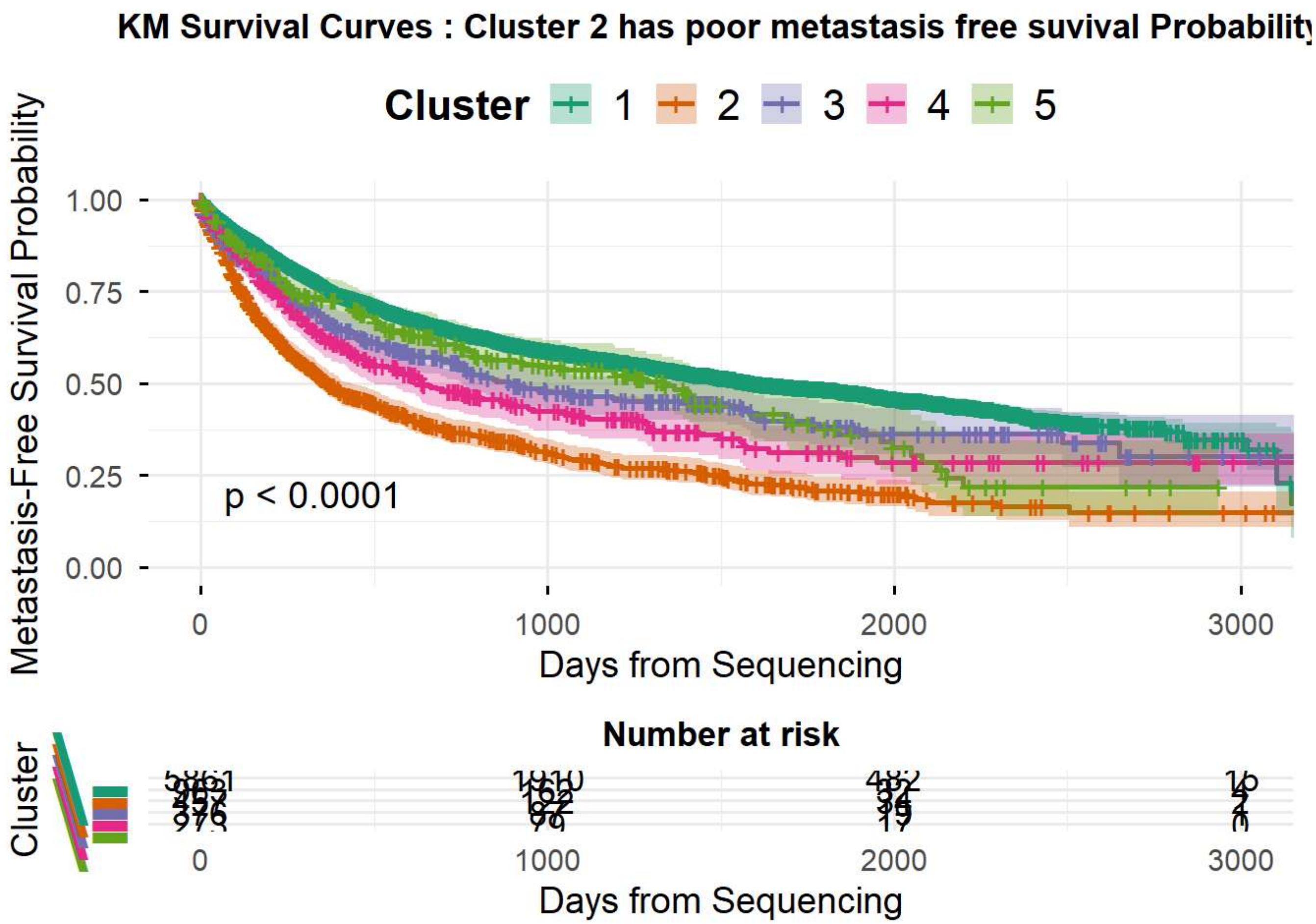
Metastasis Free Survival Analysis

Using the RSF-derived proximity matrix, Partitioning Around Medoids (PAM) identified **five distinct patient clusters**. The stability of these clusters was rigorously assessed via bootstrap validation, yielding a mean **Jaccard similarity coefficient of 0.918**, indicating highly reproducible genomic phenotypes.

```
# Validate Survival Separation

#==============================================================================
### This is a log-rank test, which tests:

###Are survival (time-to-metastasis) curves different across clusters?


#survdiff(Surv(time, event) ~ cluster,data = genomic_features)




### KM Analysis
fit <- survfit(Surv(time, event) ~ cluster,
               data = genomic_features)
km_plot <- ggsurvplot(
  fit,
  data = genomic_features,
  risk.table = TRUE,
```

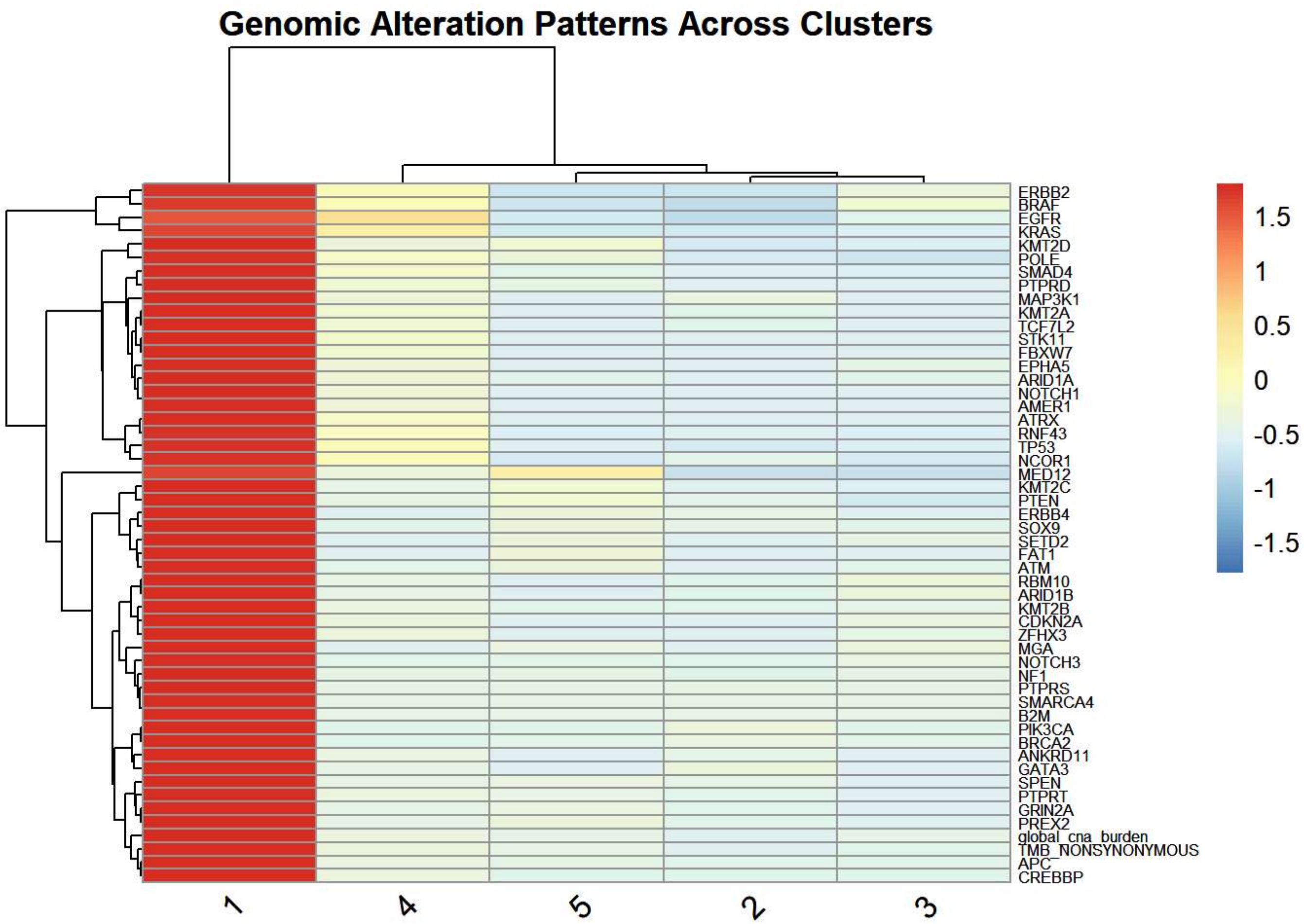
```
  pval = TRUE,
  conf.int = TRUE,
  palette = "Dark2",

  # Labels
  xlab = "Days from Sequencing",
  ylab = "Metastasis-Free Survival Probability",
  title = "KM Survival Curves : Cluster 2 has poor metastasis free suvival Probability",

  ggtheme = theme_minimal(base_size = 14) +
    theme(
      plot.title = element_text(size = 13, face = "bold", hjust = 0.5), # centered
      legend.title = element_text(size = 16, face = "bold"),
      legend.text  = element_text(size = 16)
    ),

  # Improve legend
  legend.title = "Cluster",
  legend.labs = levels(genomic_features$cluster),

  # Risk table styling
  risk.table.height = 0.25,
  risk.table.y.text.col = TRUE,
  risk.table.y.text = FALSE
)
km_plot
```

```
Ignoring unknown labels:
• colour : "Cluster"
```




```
#Heatmap of Genomic Features by Cluster(main)

#names(genomic_features)
gene_summary <- genomic_features %>%
  group_by(cluster) %>%
  summarise(across(where(is.numeric), mean))
gene_matrix <- gene_summary %>%
  dplyr::select(-time, -event) %>%
  column_to_rownames("cluster")
```

```r
heat <- pheatmap(
  t(gene_matrix),
  scale = "row",
  clustering_method = "ward.D2",
  fontsize_row = 6,
  fontsize_col = 12,

  angle_col = 45,

  main = "Genomic Alteration Patterns Across Clusters"
)

heat
```




```
#final_plot


#ggsave("genomic_cluster_plot.png",final_plot,width = 14,height = 10,dpi = 300)
```

Log-rank testing revealed highly significant differences in metastasis-free survival across the five cohorts ($p < 0.0001$).

- **Cluster 2 (High-Risk Progressors):** This group exhibited the most aggressive clinical trajectory, with an estimated **75% probability of metastasis within 500 days** post-sequencing. This phenotype is characterized by a high frequency of

TP53 mutations and *ERBB2* (HER2) alterations, alongside a significant global Copy Number Alteration (CNA) burden.

- **Cluster 4 (Indolent/Stable Phenotype):** In contrast, Cluster 4 demonstrated favorable long-term Metastais Free Survival. Despite the presence of traditionally aggressive primary cancer types, the genomic profile showed relative stability and lower overall genomic chaos suggesting an opportunity for therapeutic de-escalation.

An analysis of Cluster 1 revealed a counter-intuitive relationship between mutational density and clinical outcome. While the heatmap confirms Cluster 1 as a **hyper-mutated** , showing near-universal alterations across the gene panel, this group maintained the **highest Metasatsis Free Survival probability**

This finding aligns with the neoantigen-loadhypothesis (Goodman, A. M., et al. (2017)):

1. The elevated Tumor Mutational Burden (TMB) likely increases the production of neoantigens, rendering the tumor more susceptible to endogenous immune surveillance.

2. The extreme level of instability may impose a fitness penalty on the tumor cells, inhibiting their ability to successfully undergo the metastatic state

```
### Multivarable cox regression model and forest plot for Hazard Ratio
#prior_meds <- read.delim("msk_chord_2024_data/data_timeline_prior_meds.txt", comment.char="#")


prior_meds<- read.delim(url_prior_meds, comment.char="#",
                              stringsAsFactors = TRUE)
#head(prior_meds)

prior_meds_new <- prior_meds%>%
  rename(Prior_Meds_Binary = PRIOR_MED_TO_MSK) %>%
  mutate(Prior_Meds_Binary = ifelse(Prior_Meds_Binary == "Prior medications to MSK", 1, 0))

prior_meds_new$Prior_Meds_Binary <- as.factor(prior_meds_new$Prior_Meds_Binary)


names(clinical_features)
```

```
[1] "PATIENT_ID"                  "GENDER"
[3] "RACE"                        "ETHNICITY"
[5] "CURRENT_AGE_DEID"            "SMOKING_PREDICTIONS_3_CLASSES"
[7] "OS_MONTHS"                   "OS_STATUS"
```

```
genomic_features_full <- genomic_features%>%
  left_join(clinical_features, by = "PATIENT_ID")%>%
  left_join(prior_meds_new, by = "PATIENT_ID")

genomic_features_full$GENDER <- droplevels(genomic_features_full$GENDER)
#levels(genomic_features_full$GENDER)

#levels(genomic_features_full$cluster)




# Fit the multivariable model
# This tests: Is Cluster membership still significant AFTER accounting for Age, Sex, etc?

genomic_features_full$CANCER_TYPE <- as.factor(genomic_features_full$CANCER_TYPE)
genomic_features_full$CANCER_TYPE <- relevel(genomic_features_full$CANCER_TYPE, ref = "Prostate Cancer")
cox_model <- coxph(Surv(time, event) ~ cluster + CURRENT_AGE_DEID + GENDER +
                   Prior_Meds_Binary + CANCER_TYPE,
                   data = genomic_features_full)

# View the results
summary(cox_model)
```

```
Call:
coxph(formula = Surv(time, event) ~ cluster + CURRENT_AGE_DEID +
    GENDER + Prior_Meds_Binary + CANCER_TYPE, data = genomic_features_full)

  n= 6660, number of events= 2969
   (1270 observations deleted due to missingness)

                                             coef  exp(coef)  se(coef)      z
```

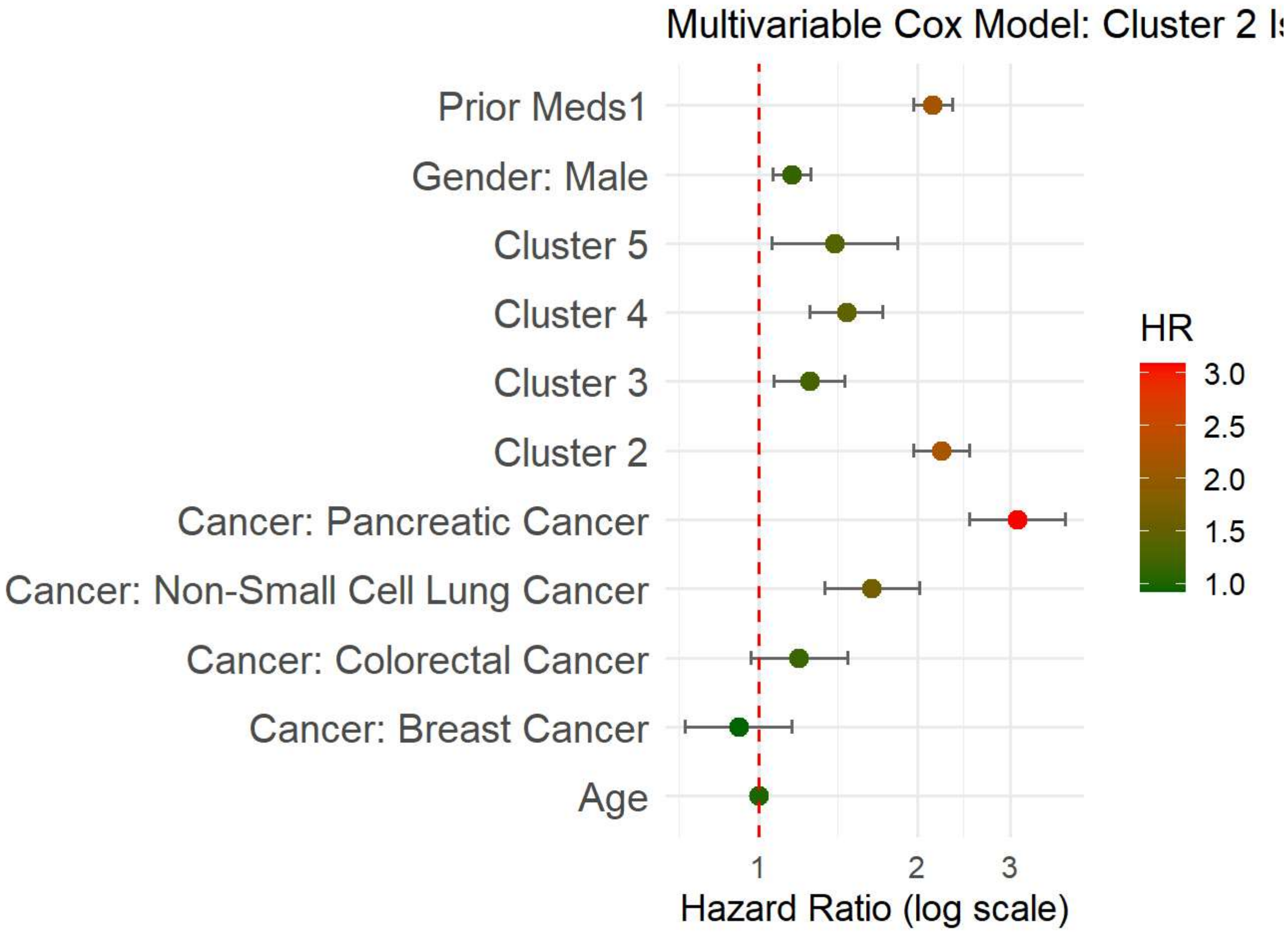
```
cluster2                             0.7983256  2.2218177  0.0619959 12.877
cluster3                             0.2220280  1.2486063  0.0787896  2.818
cluster4                             0.3823322  1.4656990  0.0811794  4.710
cluster5                             0.3331622  1.3953736  0.1403848  2.373
CURRENT_AGE_DEID                     0.0007663  1.0007666  0.0016406  0.467
GENDERMale                           0.1448860  1.1559078  0.0428569  3.381
Prior_Meds_Binary1                   0.7607895  2.1399651  0.0436853 17.415
CANCER_TYPEBreast Cancer            -0.0876503  0.9160812  0.1188274 -0.738
CANCER_TYPEColorectal Cancer         0.1769701  1.1935954  0.1085885  1.630
CANCER_TYPENon-Small Cell Lung Cancer  0.4946214  1.6398772  0.1052831  4.698
CANCER_TYPEPancreatic Cancer         1.1288081  3.0919689  0.1062968 10.619
                                    Pr(>|z|)
cluster2                             < 2e-16 ***
cluster3                            0.004833 **
cluster4                            2.48e-06 ***
cluster5                            0.017634 *
CURRENT_AGE_DEID                    0.640450
GENDERMale                          0.000723 ***
Prior_Meds_Binary1                   < 2e-16 ***
CANCER_TYPEBreast Cancer            0.460741
CANCER_TYPEColorectal Cancer        0.103158
CANCER_TYPENon-Small Cell Lung Cancer 2.63e-06 ***
CANCER_TYPEPancreatic Cancer         < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                                    exp(coef) exp(-coef) lower .95 upper .95
cluster2                               2.2218     0.4501    1.9676     2.509
cluster3                               1.2486     0.8009    1.0699     1.457
cluster4                               1.4657     0.6823    1.2501     1.718
cluster5                               1.3954     0.7167    1.0597     1.837
CURRENT_AGE_DEID                       1.0008     0.9992    0.9976     1.004
GENDERMale                             1.1559     0.8651    1.0628     1.257
Prior_Meds_Binary1                     2.1400     0.4673    1.9644     2.331
CANCER_TYPEBreast Cancer               0.9161     1.0916    0.7258     1.156
CANCER_TYPEColorectal Cancer           1.1936     0.8378    0.9648     1.477
CANCER_TYPENon-Small Cell Lung Cancer    1.6399     0.6098    1.3341     2.016
```

```
CANCER_TYPEPancreatic Cancer             3.0920     0.3234    2.5105     3.808

Concordance= 0.679  (se = 0.005 )
Likelihood ratio test= 987.8  on 11 df,   p=<2e-16
Wald test            = 1120  on 11 df,   p=<2e-16
Score (logrank) test = 1183  on 11 df,   p=<2e-16
```

```r
cox_df <- tidy(cox_model, exponentiate = TRUE, conf.int = TRUE)
cox_df <- cox_df %>%
  filter(term != "(Intercept)") %>%
  mutate(
    term = gsub("cluster", "Cluster ", term),
    term = gsub("CURRENT_AGE_DEID", "Age", term),
    term = gsub("GENDER", "Gender: ", term),
    term = gsub("Prior_Meds_Binary", "Prior Meds", term),
    term = gsub("CANCER_TYPE", "Cancer: ", term)
  )



coxplot <- ggplot(cox_df, aes(x = estimate, y = term)) +

  geom_errorbar(
    aes(xmin = conf.low, xmax = conf.high),
    orientation = "y",
    height = 0.2,
    color = "gray40"
  ) +

  geom_point(aes(color = estimate), size = 3) +

  geom_vline(xintercept = 1, linetype = "dashed", color = "red") +

  scale_x_log10() +
  scale_color_gradient(low = "darkgreen", high = "red") +

  theme_minimal(base_size = 14) +
```

```
  labs(
    title = "Multivariable Cox Model: Cluster 2 Is Significantly and Independently Associated with Higher Hazard
    x = "Hazard Ratio (log scale)",
    y = "",
    color = "HR"
  ) +
  theme(
    plot.title = element_text(size = 15),
    axis.text.y = element_text(size = 15)
  )
coxplot
```

```
`height` was translated to `width`.
```




```
#ggsave("coxHR_plot.png",coxplot,width = 10,height = 10,dpi = 300)
```

Multivariate Cox Regression Analysis

The multivariate Cox proportional hazards model was employed to evaluate the independent prognostic value of the genomic clusters while controlling for key clinical covariates ($n = 6,660$; events $= 2,969$).

Using **Cluster 1** as the reference group , all other clusters showed a significantly increased hazard of metastasis:

- **Cluster 2** remains the highest-risk genomic phenotype. With a **Hazard Ratio (HR) of 2.22** $(p < 2e - 16)$, patients in Cluster 2 are **122% more likely** to experience a metastatic event at any given time compared to Cluster 1, independent of their specific cancer type or age.
- Cluster 2 is followed by Cluster 4 (HR = 1.47), Cluster 5 (HR = 1.40), and Cluster 3 (HR = 1.25), indicating progressively elevated risk across molecular subgroups. Among clinical variables, prior medication use is a strong predictor of worse survival (HR = 2.14, p < 2e-16), suggesting substantial baseline risk differences in this group.
- Overall model performance is moderate-to-good (concordance = 0.679), and the highly significant global tests (likelihood ratio, Wald, and score tests; all p < 2e-16) confirm that the included variables collectively provide strong prognostic information for survival.

Cluster Association Analysis

To determine the biological composition of the identified subgroups, the association between molecular clusters and primary cancer types was analyzed using Pearson residuals. The analysis revealed highly significant tissue-specific enrichment

```
#Cluster Association Analysis


#table(genomic_features$CANCER_TYPE)

genomic_features_cluster <- genomic_features %>%
  mutate(CANCER_TYPE_SIMPLE = case_when(
    grepl("breast", CANCER_TYPE, ignore.case = TRUE) ~ "Breast",
    grepl("lung", CANCER_TYPE, ignore.case = TRUE) ~ "Lung",
    grepl("prostate", CANCER_TYPE, ignore.case = TRUE) ~ "Prostate",
    grepl("pancreatic", CANCER_TYPE, ignore.case = TRUE) ~ "Pancreatic",
    grepl("colorectal", CANCER_TYPE, ignore.case = TRUE) ~ "Colorectal",
    TRUE ~ "Other"
  ))
# 1. Create the contingency table
contingency_table <- table(genomic_features_cluster$cluster, genomic_features_cluster$CANCER_TYPE_SIMPLE)

# 2. Run Chi-square test
```

```r
chi_test <- chisq.test(contingency_table)
print(chi_test)
```

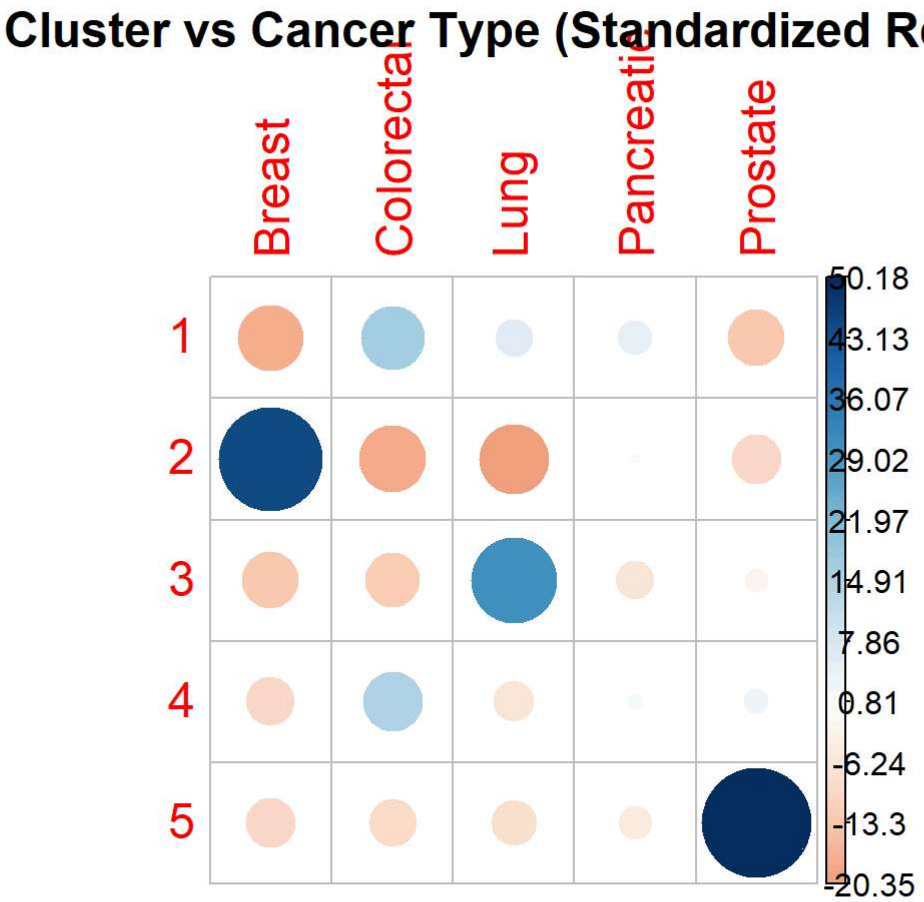
```
	Pearson's Chi-squared test

data:  contingency_table
X-squared = 5613.8, df = 16, p-value < 2.2e-16
```

```r
# 3. Extract and view standardized residuals
std_residuals <- chi_test$stdres
print(std_residuals)
```

```
         Breast  Colorectal        Lung  Pancreatic    Prostate
  1 -17.6910919  16.9696038   5.7731724   4.9820904 -13.4464234
  2  44.9365428 -18.1791891 -20.3493449  -0.4471865  -9.9424200
  3 -13.1332596 -12.4126656  30.4278572  -6.0369908  -2.2399491
  4  -9.5541846  14.6183712  -6.7935750   1.0561515   2.4736508
  5  -9.9718392  -9.4777761  -8.4219397  -4.7131579  50.1782077
```

```r
# 4. Optional: Visualize with a Mosaic Plot or Heatmap


# Use the residuals from your chi-test
#png("cluster_plot.png", width = 3000, height = 3000, res = 300)

corrplot(std_residuals,
         is.cor = FALSE,
         method = "circle",
         number.cex = 0,
         tl.col = "red",
         tl.cex = 1.5,
         cl.pos = "r",
         cl.cex = 0,
         mar = c(0,0,2,0))
```

```
title("Cluster vs Cancer Type (Standardized Residuals)",
      cex.main = 1.5,    # 👈 increase size
      font.main = 2)     # bold
```




```
#dev.off()
```

> This plot shows the association between molecular **clusters (rows)** and **cancer types (columns)** using **standardized residuals from a chi-square test of independence**. Each circle represents how much a specific cluster–cancer type pair deviates from what would be expected if there were no relationship; larger and darker circles indicate stronger deviations.
>
> **Cluster 2 and Cluster 5** demonstrated the strongest tissue specificity, being almost exclusively over-represented in **Breast Cancer** and **Prostate Cancer**, respectively. **Cluster 3** was significantly enriched for **Non-Small Cell Lung Cancer**, while **Colorectal Cancer** cases were primarily distributed between **Clusters 1 and 4**. These results indicate that while some genomic features are shared across tissues, the clustering remains strongly influenced by the primary site of the tumor.

Connecting to Survival:

- **lowest-surviving group (Cluster 2)** is the **Breast Cancer** enriched group.
- **highest-surviving group (Cluster 1)** is enriched for **Colorectal Cancer**.

It suggests that the "Breast Cancer" signature in dataset is significantly more aggressive than the "Colorectal" signature in Cluster 1.

```
### Finding High Prevalent cancer
# Count patients per cancer type
cancer_counts <- genomic_features %>%
  group_by(CANCER_TYPE) %>%
  summarise(n_patients = n()) %>%
  arrange(desc(n_patients))

kable(cancer_counts)
```

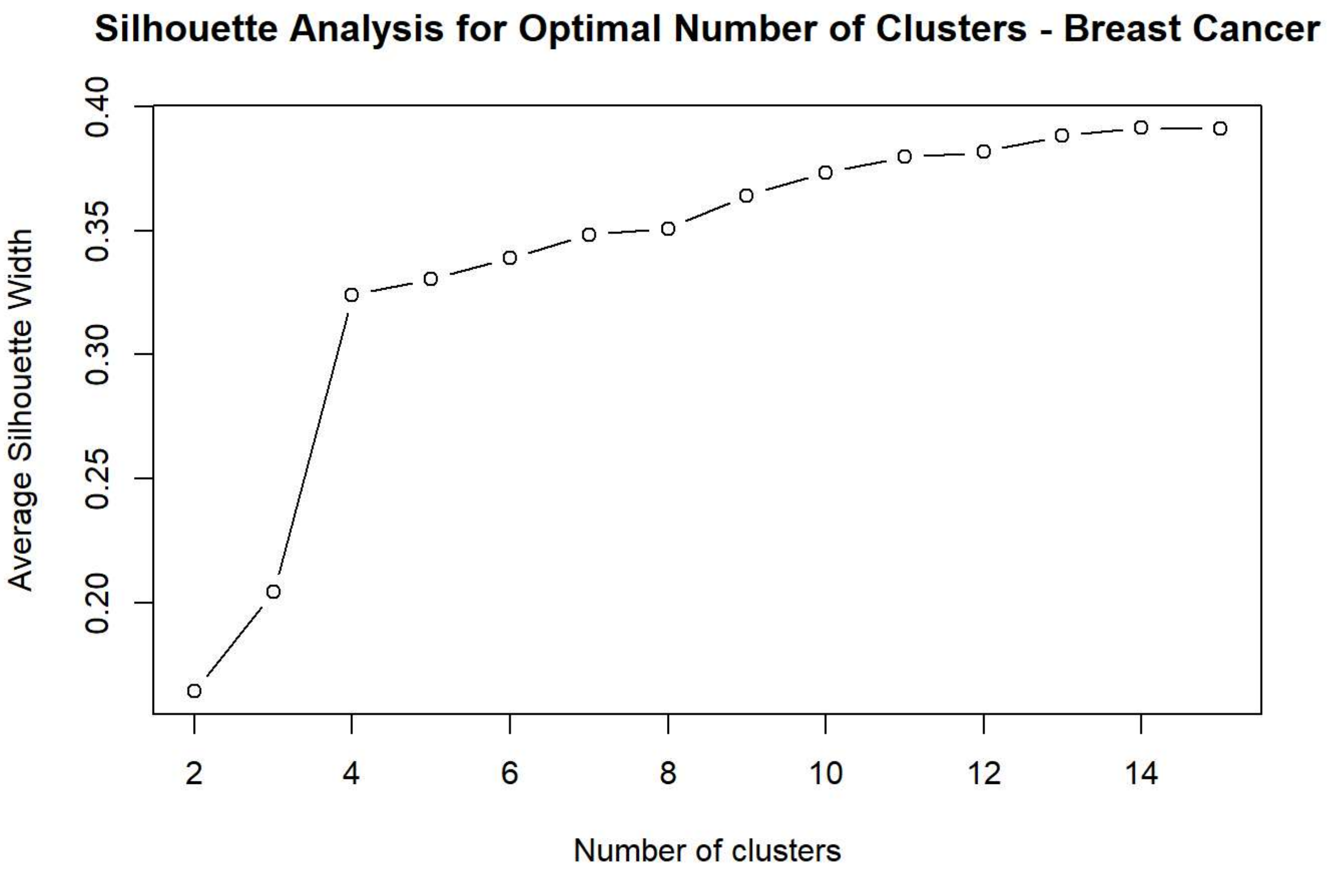
CANCER_TYPE	n_patients
Non-Small Cell Lung Cancer	2257
Breast Cancer	2106
Colorectal Cancer	1912

CANCER_TYPE	n_patients
Pancreatic Cancer	999
Prostate Cancer	656

Within type Cluster Analysis - Breast Cancer

```
### Within type RSF Clustering - Breast Cancer

breast_data <- genomic_features %>%
  filter(CANCER_TYPE == "Breast Cancer")

rsf_data <- breast_data %>%
  select(-PATIENT_ID, -SAMPLE_ID, -CANCER_TYPE)

# RSF model
set.seed(123)
rsf_model <- rfsrc(
  Surv(time, event) ~ .,
  data = rsf_data,
  ntree = 1000,
  proximity = "oob",
  importance = "TRUE"
)

# Distance matrix
dist_obj_b <- as.dist(1 - rsf_model$proximity)


# Determine optimal number of clusters (Silhouette method)
sil_widthb <- c()
for (k in 2:15) {
  pam_fit_breast <- pam(dist_obj_b, k = k, diss = TRUE)
  sil_widthb[k] <- pam_fit_breast$silinfo$avg.width
}
```

```
plot(2:15, sil_widthb[2:15], type = "b",
     xlab = "Number of clusters",
     ylab = "Average Silhouette Width")

title("Silhouette Analysis for Optimal Number of Clusters - Breast Cancer")
```




```
# Force k = 5 clustering
pam_fit_b <- pam(dist_obj_b, k = 5, diss = TRUE)
```

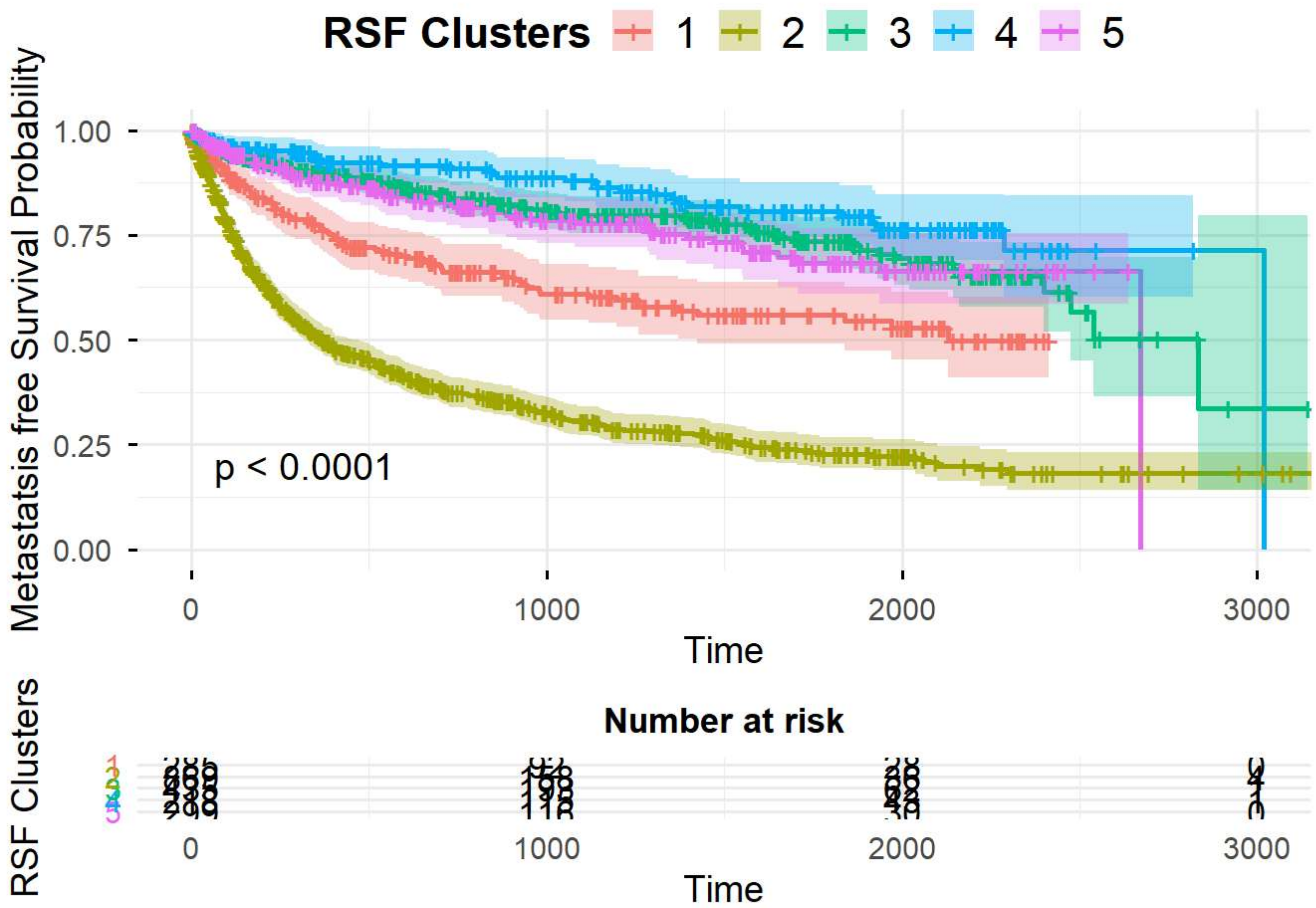
```
breast_data$cluster <- factor(pam_fit_b$clustering)
```

Although silhouette analysis reached a global maximum at K=14, K=5 was selected for the final model. This decision was driven by the need for biological interpretability; K=5 corresponds to the five well-characterized intrinsic molecular subtypes of breast cancer. Furthermore, K=5 represents a critical point where cluster cohesion significantly improves without leading to the excessive fragmentation or overfitting associated with higher K values.

Metastasis Free Survival Analysis for Breast Cancer

```
km_fit_breast <- survfit(Surv(time, event) ~ cluster, data = breast_data)
km_plot_breast <- ggsurvplot(
  km_fit_breast,
  data = breast_data,
  risk.table = TRUE,
  pval = TRUE,
  conf.int = TRUE,

  xlab = "Time",
  ylab = "Metastatsis free Survival Probability",

  ggtheme = theme_minimal(base_size = 14) +
    theme(
      plot.title = element_text(size = 13, face = "bold", hjust = 0.5),
      legend.title = element_text(size = 16, face = "bold"),
      legend.text  = element_text(size = 16)
    ),

  legend.title = "RSF Clusters",
  legend.labs = levels(breast_data$cluster)
)

km_plot_breast
```

```
Ignoring unknown labels:
• colour : "RSF Clusters"
```



Kaplan-Meier (KM) Survival Curves

The plot on the left illustrates Metastasis-free Survival Probability over time for five distinct patient clusters in breast cancer patients

Significant Differentiation: The $p < 0.0001$ indicates a highly statistically significant difference in survival between the clusters.

Best vs. Worst Outcomes: * Cluster 4 (Blue): Shows the highest survival probability, maintaining a high plateau for the longest duration.

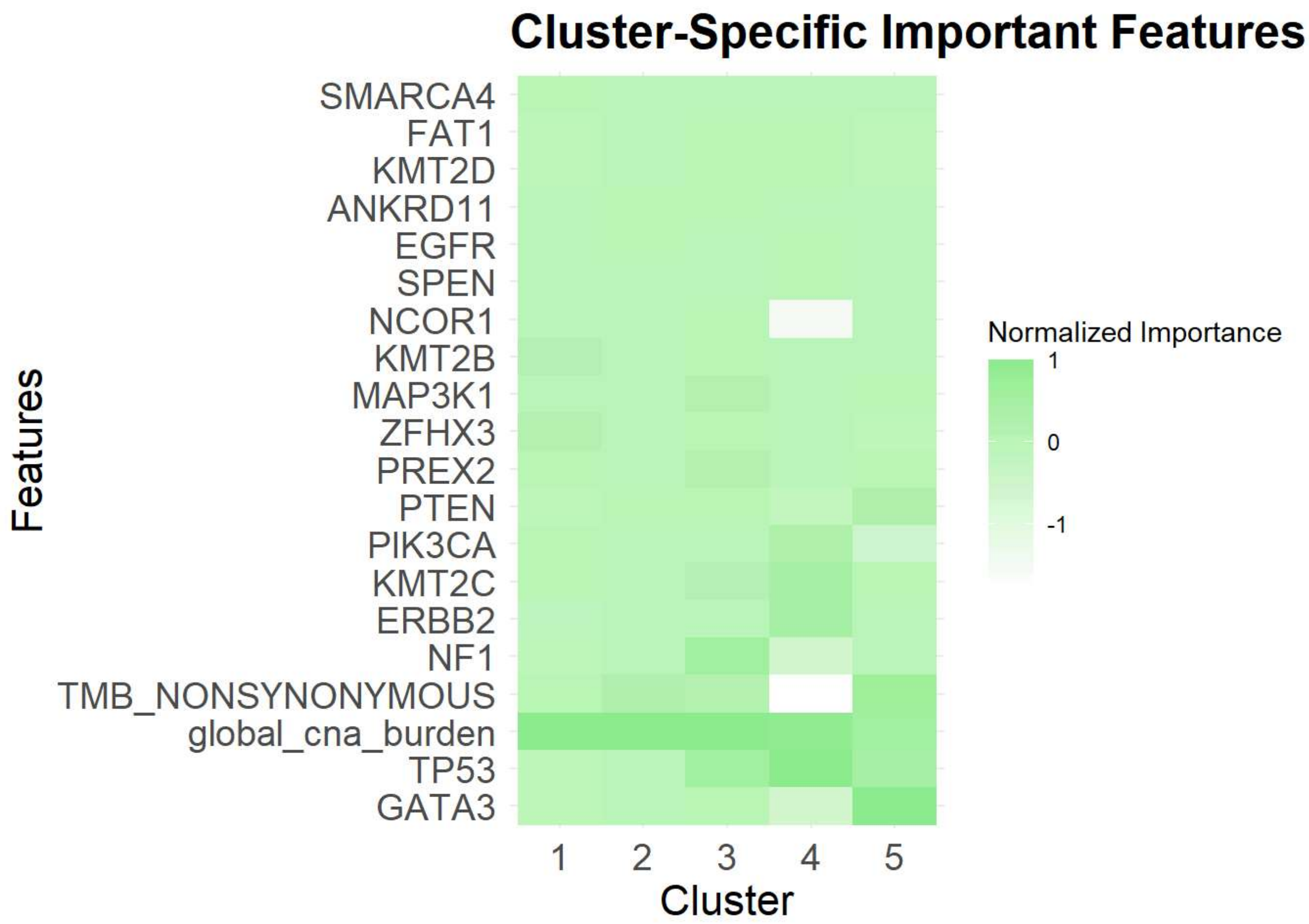
Cluster 2 (Olive Green): Represents the highest risk group, with survival probability dropping sharply and significantly faster than all other groups.It demonstrates an extremely aggressive clinical course, with overall survival probability dropping below 25% significantly faster than all other cohorts ($p < 0.0001$)

Intermediate Groups: Clusters 1 (Red), 3 (Green), and 5 (Pink) show moderate survival rates, with Cluster 3 appearing to have better long-term outcomes than Clusters 1 and 5

```r
### Breast Cancer Feature Importance

library(randomForestSRC)
library(dplyr)

vip_list <- list()

for (cl in unique(breast_data$cluster)) {

  cluster_data <- breast_data %>%
    filter(cluster == cl) %>%
    select(-PATIENT_ID, -SAMPLE_ID, -CANCER_TYPE, -cluster)

  set.seed(123)
  rsf_fit <- rfsrc(
    Surv(time, event) ~ .,
    data = cluster_data,
    ntree = 500,
    importance = "permute"
  )

  vip <- data.frame(
    variable = names(rsf_fit$importance),
    importance = rsf_fit$importance,
    cluster = cl
  )
```

```r
  vip_list[[as.character(cl)]] <- vip
}

vip_df <- bind_rows(vip_list)

vimp_df <- vip_df %>%
  group_by(cluster) %>%
  mutate(norm_importance = importance / max(importance)) %>%
  ungroup()


heatmap_df <- vimp_df %>%
  select(variable, cluster, norm_importance) %>%
  pivot_wider(
    names_from = cluster,
    values_from = norm_importance,
    values_fill = 0
  )

heatmap_long <- heatmap_df %>%
  pivot_longer(
    -variable,
    names_to = "cluster",
    values_to = "importance"
  )

top_vars <- heatmap_long %>%
  group_by(variable) %>%
  summarise(max_imp = max(importance, na.rm = TRUE)) %>%
  arrange(desc(max_imp)) %>%
  slice_head(n = 20) %>%
  pull(variable)

heatmap_long <- heatmap_long %>%
  filter(variable %in% top_vars)
```

```r
var_order <- heatmap_long %>%
  group_by(variable) %>%
  summarise(max_imp = max(importance, na.rm = TRUE)) %>%
  arrange(desc(max_imp)) %>%
  pull(variable)

heatmap_long$variable <- factor(heatmap_long$variable, levels = var_order)

heat_breast <- ggplot(
  heatmap_long,
  aes(
    x = cluster,
    y = variable,
    fill = importance
  )
) +
  geom_tile() +
  scale_fill_gradient(low = "white", high = "lightgreen") +
  theme_minimal() +
  labs(
    title = "Cluster-Specific Important Features (RSF)",
    x = "Cluster",
    y = "Features",
    fill = "Normalized Importance"
  )+
  theme(
    axis.text.x = element_text(size = 14),
    axis.text.y = element_text(size = 14),
    axis.title.x = element_text(size = 16),
    axis.title.y = element_text(size = 16),
    plot.title = element_text(size = 18, face = "bold")
  )

heat_breast
```



The heatmap identifies which features the RSF model relied on most to predict these survival outcomes across clusters:

Dominance of Global Drivers: global_cna_burden (Copy Number Alterations) and TP53 mutations emerge as universally important features across almost all clusters, particularly in the high-risk groups (Clusters 1, 2, and 5).

Cluster-Specific Drivers:

- Cluster 5: Shows a uniquely high reliance on GATA3 and TP53, suggesting these mutations are the primary determinants of this group's specific survival path.
- Cluster 3: Shows a specific reliance on NF1, distinguishing its genomic survival profile from the others.

- TMB : TMB_NONSYNONYMOUS (Tumor Mutational Burden) shows varying importance, being highly relevant for Cluster 1 and Cluster 5, but notably less so for the stable Cluster 4 (as indicated by the white/missing block).

The aggressive nature of Cluster 2 is largely explained by a high CNA burden and TP53 mutations. Conversely, the model's reliance on different genes (like NF1 for Cluster 3 vs. GATA3 for Cluster 5) to predict similar survival curves suggests that different genomic "paths" can lead to the same clinical outcome.

Conclusion

This research demonstrates that unsupervised clustering on RSF-derived distances effectively reveals hidden, clinically relevant patient subtypes. This approach bridges the gap between genomic complexity and survival forecasting, offering a robust pipeline for precision medicine and the identification of high-risk patients who may benefit from intensified therapeutic interventions.

References

1.Filipp, F. V., et al. (2017). Precision medicine driven by cancer genomics: implications for metastasis and clinical outcomes. Cancer Metastasis Reviews, 36(1), 91–108.
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